
Homework 1

— Music Era and Release-Market
Classification —

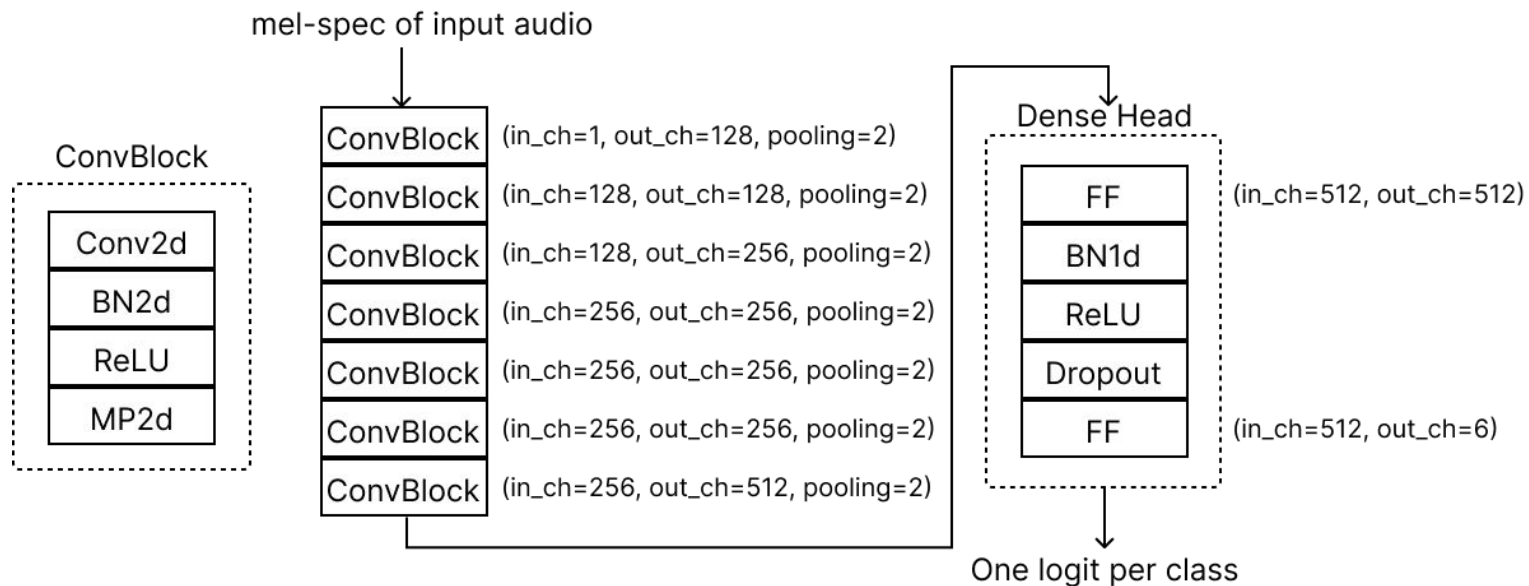
b12901045 吳岳霖

Files available at [DeepMIR_2026fall/hw1](https://deepmir.github.io/2026fall/hw1)

Outline

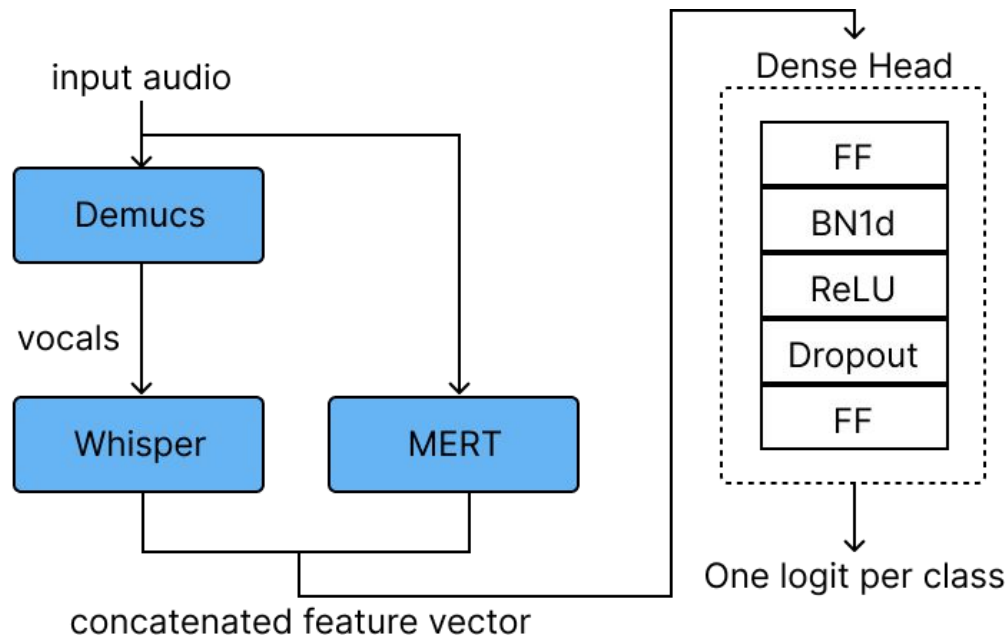
- Methods:
 - Short-Chunk CNN
 - Source Separation + Pretrained models + Dense classifier
- Results
- Insights
- Future work
- References

Methods: Short-Chunk CNN



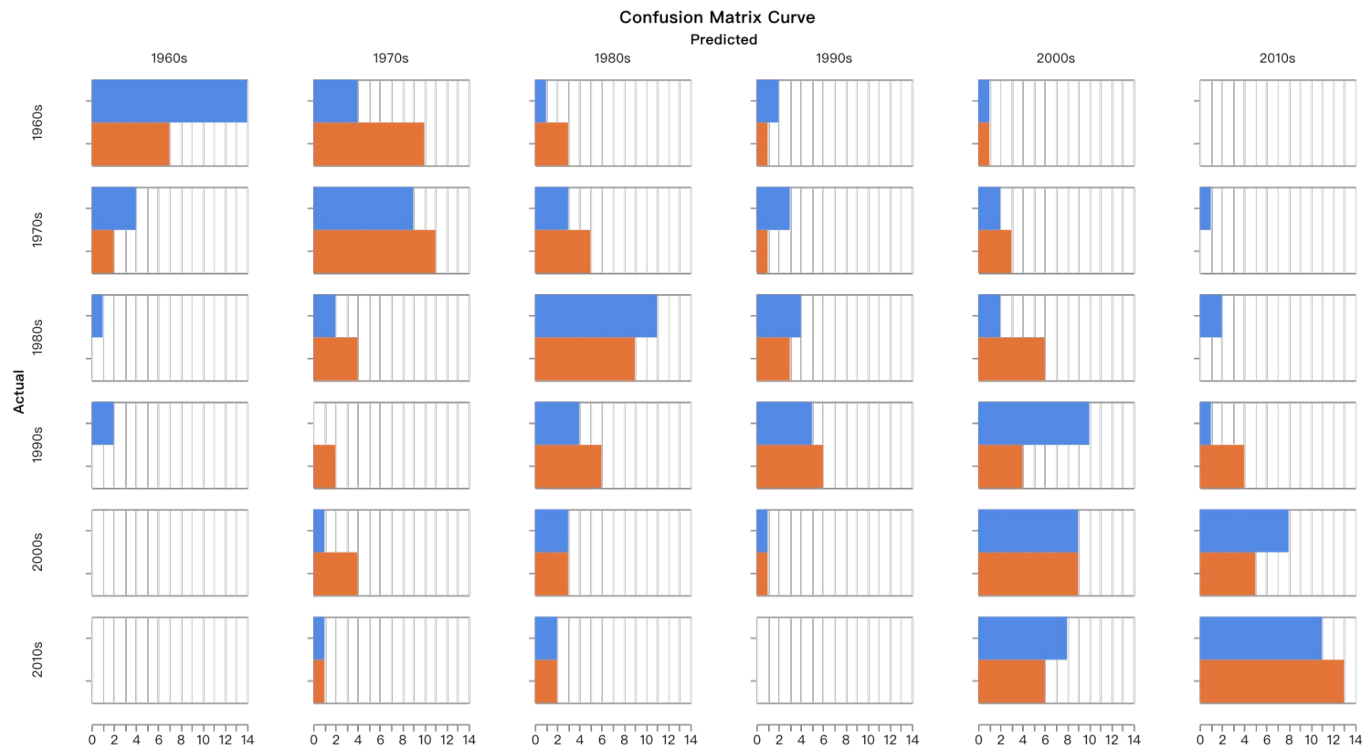
- A trained-from-scratch short chunk CNN model with 3 chunks of 4 seconds from each clip.
- Reference from [Evaluation of CNN-based Automatic Music Tagging Models](#)

Methods: Pretrained models + Dense Head



- A small trained dense classifier that takes the concatenated feature vector from [Whisper](#) and [MERT](#).
- Different layer's outputs are weighted by a learned parameter before concatenating.
- For task B, the raw probability of the audio's language is concatenated too.

Results - Task A



- Blue: Short-Chunk CNN
- Orange: Pretrained + Dense

	Top1	Top3
Blue	0.48	0.84
Orange	0.44	0.74

Results - Task B



- Green: Short-Chunk CNN
- Red: Pretrained + Dense

	Top1	Top3
Green	0.33	0.72
Red	0.59	0.86

Insights - Task A

- Errors do really happen a lot in consecutive years. 2010s is the most obvious that most of the output predicts the result as 2000s.
- For 1990s data, its accuracy is relatively lower than other release decades. After listening to some test samples, I did find a good portion of data that resembles other decades of music, which I think is reasonable that both models perform worse.
- Short-Chunk CNN performs better than the pretrained features with dense head even though they share the same backend (same dense head), which isn't what I expected.
 - I think since CNN is taking chunks of the audio input, it essentially has more data. With less data, although the features extracted by MERT and Whisper are more powerful, the dense classifier easily overfit and performs inferiorly. The assumption will need further experiments to verify.

Insights - Task B

- UK and Germany have the lowest accuracy. I think it is reasonable to mistake UK as US. However, after checking the audio, Germany market does consist of a good portion of non-German languages, which is the reason causing the confusion to the models.
- In hope that most of the markets sing with the native language of the country, I did source separation with [Demucs](#) to separate vocals and non-vocals. Inspired by my friend's method, I decided to use Whisper to extract the vocal's feature and MERT to extract the original audio's feature.
- The pretrained features + dense classifier method wins significantly compared with Short-Chunk CNN. I think the release market does have high relevance to the language of the songs.
 - I also read from [END-TO-END LEARNING FOR MUSIC AUDIO TAGGING AT SCALE](#) that when data is scarce, methods utilizing domain knowledge may perform better than those deep learning methods.

Future Work

- Different chunk length for Short-Chunk CNN and other data augmentation method: I think the challenge with this task is the scarcity of data, so data augmentation might help improve the model's performance a lot for both Short-Chunk CNN model and pretrained feature + dense classifier.
- Inspect the actual contribution of the feature from Whisper and MERT: My friend's method is feeding the original audio to Whisper while mine, inspired by his, is feeding vocals to Whisper since I think vocal is closer to the domain where it's trained on, but this suggestion needs further experiments to verify. Experiments comparing the contribution of the features from MERT and Whisper are also worth studying.

References

- END-TO-END LEARNING FOR MUSIC AUDIO TAGGING AT SCALE: <https://arxiv.org/pdf/1711.02520>
- Evaluation of CNN-based Automatic Music Tagging Models: <https://arxiv.org/abs/2006.00751>
- Demucs repo: <https://github.com/adefossez/demucs>
- Whisper: <https://arxiv.org/pdf/2212.04356>
- MERT: <https://arxiv.org/pdf/2306.00107>